

Fundamentals of Artificial Intelligence and Knowledge Representation – Module 4

Prof. Federico Chesani – 11th of January, 2021

Available time: 25 min. + 10 min. for submitting the essay

Notice: when submitting your essay, please name the file in the following way:

FamilyNameFirstName.date.pdf/.docx

For example, if you are John Smith, the file will be named:

SmithJohn.20200111.pdf

Exercise

- 1) The candidate is invited to shortly introduce rule-based forward reasoning, and to briefly highlight the main difference w.r.t. backward reasoning.
- 2) The candidate is invited to write a meta-interpreter **verbose(Goal)** that is evaluated to true if **Goal** can be proved; moreover, before trying to solve any sub-goal, the meta-interpreter must print (at the standard output) the sub-goal itself.

For example, given the program:

```
p(X) :- q(X), r(X).
```

```
p(X) :- s(X).
```

```
q(X) :- t(X).
```

```
r(1).
```

```
r(2).
```

```
r(3).
```

```
t(1).
```

```
t(2).
```

and the query `?- verbose(p(X))`, the following outcomes are expected:

```
?- solve(p(X)).
```

```
Solving: p(_4250)
```

```
Solving: q(_4250)
```

```
Solving: t(_4250)
```

```
Solving: r(1)
```

```
X = 1 ;
```

```
Solving: r(2)
```

```
X = 2 ;
```

```
Solving: s(_4250)
```

```
false.
```

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Solution

1) *The candidate is invited to shortly introduce backward and forward reasoning, and to briefly highlight the differences between the two paradigms.*

It is expected that the candidate mentions the main differences between forward and backward reasoning.

2) *The candidate is invited to write a meta-interpreter `verbose(Goal)` that is evaluated to true if `Goal` can be proved; moreover, before trying to solve any sub-goal, the meta-interpreter must print (at the standard output) the sub-goal itself.*

```
verbose(true) :- !.  
verbose(A,B) :- !, solve(A), solve(B).  
verbose(A) :-  
    write('Solving: '), write(A), nl,  
    clause(A,B),  
    verbose(B).
```